F79MB- Project 1-2026

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1. (a) A histogram, boxplot, and numerical summaries were used to analyze the distribution of bug resolution times.

Min.	1st Qu	Median	Mean	3rd Qu.	Max.
29.00	69.75	95.00	108.55	123.50	560.00

Table 1. Numerical summary of bug resolution times

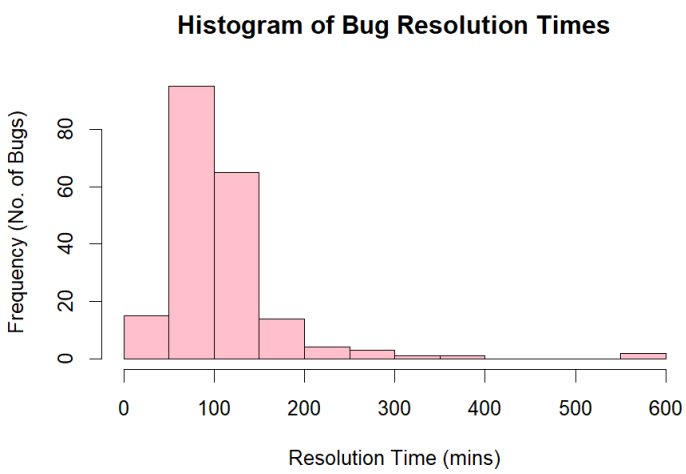

Figure 1. Histogram of Bug Resolution times

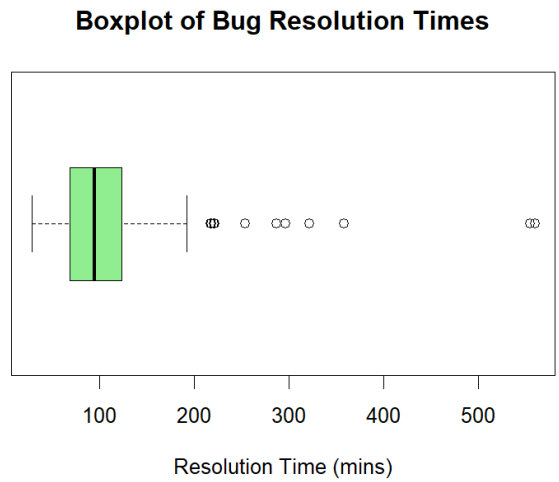

Figure 2. Boxplot of Bug Resolution times

The sample standard deviation is 68.12 minutes.
From table 1, the minimum and maximum resolution times were 29 and 560 minutes, indicating a wide range. The sample mean is higher than the median which leads to positive skewness that is further confirmed by the histogram and boxplot shown in figure 1 and 2.
The standard deviation is relatively large compared to the mean, indicating dispersion in resolution times which may be driven by extreme high values.
The histogram in figure 1 shows a very positive skewed distribution since the mean is greater than the median, and the distribution appears to be unimodal. In figure 2, the boxplot confirms this skewness through a longer upper whisker and a lot of outliers after resolution time passes 200 minutes.
Overall, the distribution is positively skewed with extreme high values, suggesting that a transformation (such as the logarithm) may be appropriate for modelling.

(b) (i) The natural logarithm of the bug-fix resolution time may be modelled by a normal distribution with parameters μ and σ^2. The data where Y_i is the natural log of the bug resolution times follows: $Y_i = \log(X_i) \sim \mathcal{N}(\mu, \sigma^2)$ where X_i are the bug resolution times. Since it's a normal distribution Y_i has the following pdf: $f(y_i) = \frac{1}{\sqrt{2\pi\sigma^2}}\, exp\left(-\frac{(y_i-\mu)^2}{2\sigma^2}\right)$.

Assuming Y_i are independent observations we can take the product of it for the likelihood:

$L(\mu,\sigma^2) = \prod_{i=1}^{n} f(y_i) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}}\, exp\left(-\frac{(y_i-\mu)^2}{2\sigma^2}\right) = (2\pi)^{-\frac{n}{2}}(\sigma^2)^{-\frac{n}{2}} \prod_{i=1}^{n} exp\left(-\frac{(y_i-\mu)^2}{2\sigma^2}\right)$

For the log-likelihood, take log of both sides: $\ell(\mu,\sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i-\mu)^2$.

To find the MLE for μ differentiate with respect to μ which gives the score function:

$$\frac{\partial \ell}{\partial \mu} = -\frac{1}{2\sigma^2}\cdot 2\sum_{i=1}^{n}(y_i-\mu)(-1) = \frac{1}{\sigma^2}\sum_{i=1}^{n}(y_i-\mu).$$

Equating this score function to zero:

$$\sum_{i=1}^{n}(y_i-\mu) = 0 \Rightarrow \hat{\mu} = \frac{\sum_{i=1}^{n} y_i}{n} = \bar{y}.$$

To find the MLE for σ^2 differentiate with respect to σ^2 which gives the score function:

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2}\cdot\frac{1}{\sigma^2} + \frac{1}{2}\cdot\frac{1}{(\sigma^2)^2}\sum_{i=1}^{n}(y_i-\mu)^2.$$

Equating this score function to zero:

$$-\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(y_i-\hat{\mu})^2 = 0$$

$$\frac{\sum_{i=1}^{n}(y_i-\hat{\mu})^2}{2(\sigma^2)^2} = \frac{n}{2\sigma^2} \;\Rightarrow\; \frac{\sum_{i=1}^{n}(y_i-\hat{\mu})^2}{n} = \frac{2(\sigma^2)^2}{2\sigma^2}$$

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{n}(y_i-\hat{\mu})^2}{n}$$

Hence, the MLEs are:

$$\hat{\mu} = \bar{y}\,, \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i-\bar{y})^2.$$

(b)(ii) Using log-transformed data $y_i = \log(x_i)$, the MLE of μ is $\hat{\mu} = \frac{\sum_{i=1}^{n} y_i}{n}$. From the data, this gives $\hat{\mu} = 4.563369$.

Computing the sample mean using R directly also gives $\bar{x} = 4.563369$.

The MLE of the variance is $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i-\bar{y})^2$. Using the given data, $\hat{\sigma}^2 = 0.2187059$.

Computing the sample variance using R directly is $s^2 = 0.2198049$.

The MLE of μ is identical to the sample mean of the log-transformed data since both divides by n. However, the MLE of σ^2 is slightly smaller than the sample variance. This is because the MLE divides by n whereas the sample variance divides by n − 1. The difference between the two quantities is very small as the sample size for the bug data is 200. Larger the sample size, difference becomes more negligible.

(c)

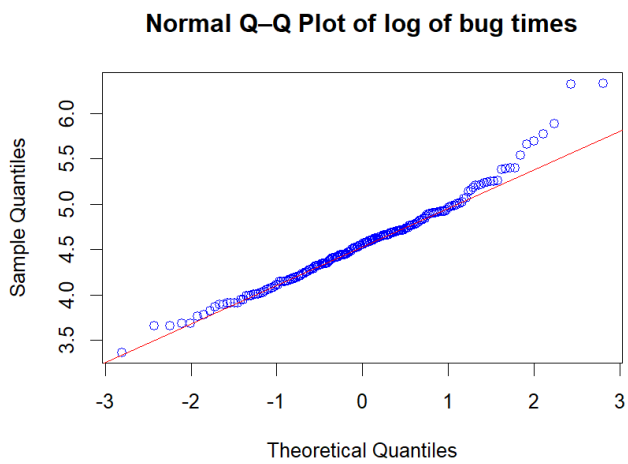

Figure 3. Normal Q-Q Plot of log of bug times

Analyzing figure 3, majority of the points lie close to the reference line, especially in the central area of the distribution. This suggests that the log-transformed data are approximately normally distributed. However, there is an evident upward deviation in the upper tail, where several observations lie above the line. This indicates that the right tail is a bit heavier than expected under a normal distribution. Overall, the plot suggests that the normal distribution provides a reasonable approximation for the log-transformed data, although some tail deviations remain.

(d) Performing a chi-square goodness-of-fit test where the hypotheses test:

H_0: The bug resolution times follow a Lognormal distribution
H_1: The bug resolution times do not follow a Lognormal distribution.

Taking the parameter estimates from part (b)(ii), the fitted lognormal distribution was divided into 10 equal-probability intervals using its quantiles. Since each interval has probability 0.1 under H_0 and the sample size is $n = 200$, the expected frequency in each cell is: $e_i = 200 \times 0.1 = 20, i = 1,2, \dots ,10$. The observed frequencies along with its corresponding equal-probability intervals are shown in the table below.

Interval (minutes)	Observed frequency
[0, 52.7]	17
(52.7, 64.7]	22
(64.7, 75]	22
(75, 85.2]	23
(85.2, 95.9]	17
(95.9, 108]	24
(108, 123]	24
(123, 142]	19
(142, 175]	12
(175, ∞]	20

Table 2. Observed frequencies for 10 equal-probability intervals under the fitted lognormal model.

The chi-square test statistic is $\chi^2 = \sum_{i=1}^{10} \frac{(f_i - e_i)^2}{e_i} = 6.6$, when computed by R. Having 10 cells and estimated parameters μ and σ^2 , the test statistic χ^2 has df = 10-1-2 = 7. Hence p-value = $P(X\ \frac{2}{7} > 6.6)$ = 0.4716799. Since this p-value is greater than 0.05, there is no sufficient evidence to reject the null hypothesis at the 5% significance level. The data does not show a significant difference from the fitted lognormal model. The results are consistent with the assumption that the bug resolution times follow a lognormal distribution.

2.(a) A histogram, boxplot, and numerical summaries were used to analyze the distribution of a sample of 32 app session durations (in minutes).

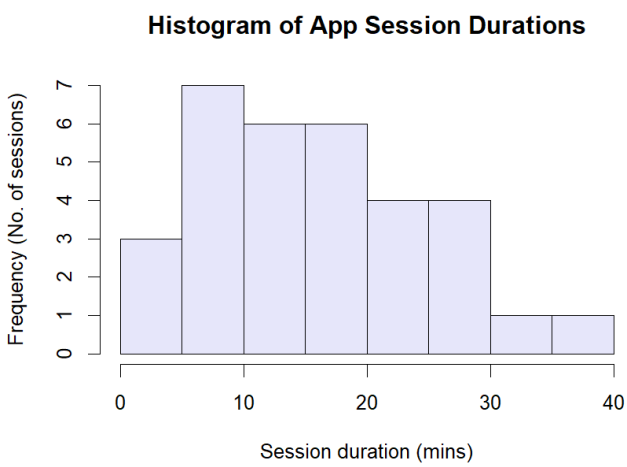

Figure 4. Histogram of App Session Durations

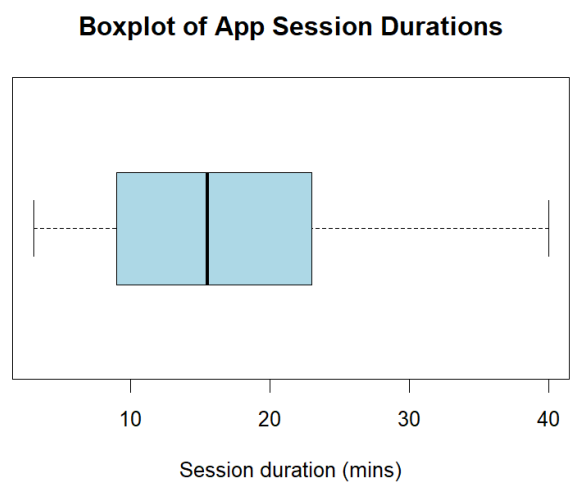

Figure 5. Boxplot of App Session Durations

Min.	1st Qu	Median	Mean	3rd Qu.	Max.
3.00	9.00	15.50	16.62	22.50	40.00

Table 3. Numerical summaries for App durations

The sample standard deviation is 9.328 minutes.
From table 2, the minimum and maximum session durations were 3 and 40 minutes, indicating a wide range of usage times. The sample mean is larger than the median which results in a small right skew in the distribution. The standard deviation of 9.33 minutes represents more than half of the mean, indicating lots of variability in session durations.
The histogram in figure 4 shows that most session lengths lie between approximately 5 and 20 minutes. The distribution appears right-skewed, confirming the above statement due to the mean being larger than the median.
The boxplot in figure 5 supports this too. The upper whisker is longer than the lower whisker, and the median lies slightly below the centre of the box, both suggesting positive skewness. No extreme outliers are evident.
Because the sample size is relatively small (n = 32) and the distribution shows some skewness, inference methods that depend strongly on normality may be less reliable. For this reason, non-parametric approaches such as bootstrap methods are appropriate for estimating quantities like the median and computing confidence intervals, since they rely less on distributional assumptions.

(b)

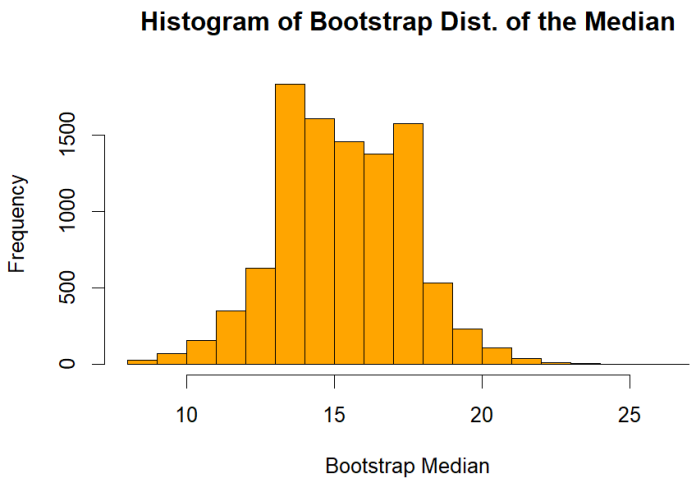

Figure 6. Histogram of Empirical Bootstrap Sampling Distribution of

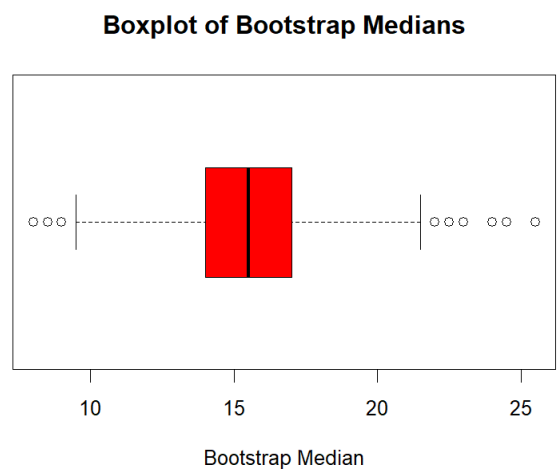

Figure 7. Boxplot of Empirical Bootstrap Sampling Distribution of

Min.	1st Qu	Median	Mean	3rd Qu.	Max.
8.00	14.00	15.50	15.57	17.00	25.50

Table 4. Numerical summaries for Bootstrap distribution of median

A non-parametric bootstrap method was used to check the variability of the sample median. From the original sample of 32 session durations, 10,000 resamples of size 32 were drawn with replacement, and the median was calculated for each sample to approximate its sampling distribution.

The bootstrap histogram shows a roughly symmetric and unimodal distribution centred near the observed median of 15.5 minutes. While the original data exhibited some skewness, the bootstrap distribution appears approximately symmetric and bell-shaped resembling a normal distribution. The boxplot indicates moderate spread, with most bootstrap medians between approximately 14 and 17 minutes. Although it has many outliers in the figure.

Confidence interval at 95% for the median was obtained from the 2.5th and 97.5th percentiles of the bootstrap distribution: $(11.0,\ 19.5)$.

This interval suggests that the true median session duration is likely to lie within this range, and the relatively concentrated bootstrap distribution indicates that the sample median provides a stable estimate of the population median, even with a sample size of 32.

(c) Assuming session durations follow a lognormal distribution, the log-transformed data were treated as normally distributed. Under asymptotic theory, the sample median satisfies $\hat{m} \sim N(m, \left(\frac{1}{2f(m)\sqrt{n}}\right)^2)$ where $f(m)$ is the density of the Normal distribution evaluated at m and $\hat{m}$ is the sample median under lognormal model. From the data we have

$\hat{m} = 2.7403, \hat{\sigma} = 0.6486, n = 32$. Hence $f(m) = \frac{1}{\hat{\sigma}\sqrt{2\pi}} \approx 0.6152$ with estimate of standard error of median, $\text{se}(\hat{m}) = \frac{1}{2f(m)\sqrt{n}} = 0.1437$.

A 95% confidence interval for the population median on the log-scale is

$$\hat{m} \pm 1.96 \times \text{se}(\hat{m}),$$

$$\Rightarrow (2.4587,\ 3.0220).$$

Since the median of a lognormal distribution equals $\exp(m)$, exponentiating the interval gives the 95% confidence interval for the population median session duration: $(11.69,\ 20.53)$.

Comparing the asymptotic interval to the non-parametric bootstrap confidence interval obtained in part (b), it can be concluded that the asymptotic confidence interval $(11.69,\ 20.53)$ is slightly wider than the non-parametric bootstrap interval obtained in part (b), $(11.0,\ 19.5)$. The two intervals are reasonably close, suggesting that the parametric and non-parametric approaches both lead to similar conclusions about the population median.
The asymptotic interval is derived under the assumption that the log-transformed data follow a normal distribution and relies on large-sample approximations. Since the sample size is relatively small ($n = 32$) and the original data show mild skewness, the bootstrap method may provide a more reliable estimate because it does not depend on distributional assumptions. However, the similarity between the two intervals suggests that the lognormal model provides a reasonable description of the data.

(d) A total of 10,000 samples of size 32 were drawn with replacement, and the IQR was computed for each resample to approximate its sampling distribution. Confidence interval at 95% for the IQR is $(8.25,\ 18.25)$. This interval is wider than the bootstrap confidence interval for the median, indicating that the spread of session durations is more variable than their central location. In other words, while the typical session length is stable, the variability in session durations fluctuates more, likely influenced by some longer sessions.

Compared to the bootstrap confidence interval for the median obtained in part 2(b), this interval is wider, indicating greater uncertainty in estimating the spread than the center. This suggests that while the typical session duration is relatively stable, the session lengths variability is more likely prone to fluctuate the data. This increased variability in the IQR is due to some longer sessions in the sample.

(e) The app session duration data follows the lognormal distribution in 2(c). A parametric bootstrap hypothesis test is done to assess whether the population median app session duration is 15 minutes or less.

Hypothesis test:

H_0 : Median ≤ 15
H_1 : Median > 15

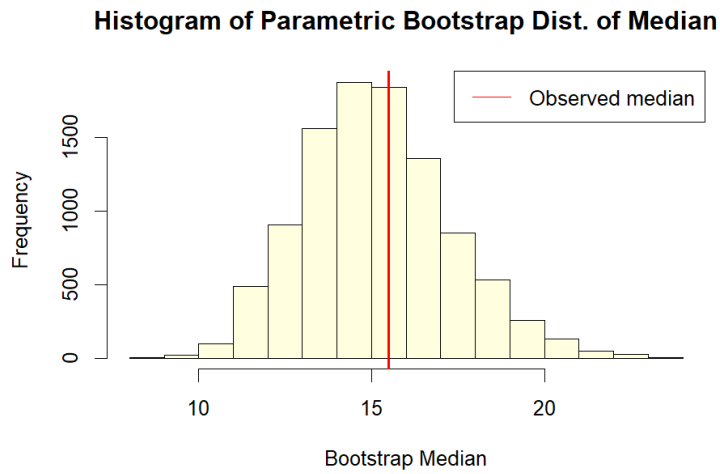

Figure 8. Histogram of Parametric Bootstrap Distribution of median

For a lognormal distribution, the median equals e^{μ}. Hence, we have $\mu_0 = \log(15)$ under H_0. Using the estimated value of σ from part (c), 10,000 samples of size 32 were simulated from a lognormal distribution with parameters μ_0 and $\hat{\sigma}$. The median was computed for each simulated sample to generate the null distribution.

The test statistic in this case is the observed sample median which is 15.5 minutes. The bootstrap p-value = 0.4045, that was calculated as the proportion of simulated medians greater than or equal to 15.5.

The histogram of the medians in figure 8 shows that the observed median that is highlighted with a vertical red line is in the central area of the null distribution rather than in the extreme right tail. Additionally, since p-value is larger than 0.05, there is insufficient evidence to reject the null hypothesis at 5% significance level. The data does not provide enough evidence that the population median session duration exceeds 15 minutes.

Appendix : R code

```
# Project 1 --- H00453164 -- Fathima Aushat Azaar

# Question 1

# Q1a
# Load the data
bug_time = scan("C:/HWU/Year 3/SEM 2/Stats Model B/cw1/bug_time.txt") ; bug_time

# Histogram and boxplot to visualize the data
hist(bug_time, main = "Histogram of Bug Resolution Times",
    xlab = "Resolution Time (mins)",
    ylab = "Frequency (No. of Bugs)",col = "pink")

boxplot(bug_time, main = "Boxplot of Bug Resolution Times",
       xlab = "Resolution Time (mins)",
       col = "lightgreen", horizontal = TRUE)

# Numerical summaries for the data
summary(bug_time)
sd(bug_time)      # sample standard deviation
var(bug_time)      # sample variance

# Q1b(ii)

log_bug = log(bug_time)  # log-transform of the data

# MLE of mu
mu_hat = sum(log_bug)/length(log_bug)

# Sample mean
sample_mean = mean(log_bug) ; mu_hat

# MLE of var
n = length(log_bug); n  # sample size
sigma2_hat = sum((log_bug - mu_hat)^2) / n ; sigma2_hat

# Sample variance
sample_var = var(log_bug) ; sample_var  # R variance uses /(n-1)


# Q1c

# Q-Q plot against Normal
qqnorm(log_bug, main = "Normal Q–Q Plot of log of bug times", col = "blue")  # to check normality of log-data
qqline(log_bug, col = "red")  # reference line of the normal distribution


# Q1d

# Chi-square goodness of fit test
```

```
ncell = 10 # 10 cells

# Estimate parameters from log-data
mu_hat = mean(log_bug)   # mean
sigma2_hat = sum((log_bug - mu_hat)^2) / n  # variance
sigma_hat = sqrt(sigma2_hat)     # standard dev

# 10 equal-probability cells under fitted lognormal
probs = seq(0, 1, length.out = 11)

bug.breaks = qlnorm(probs, mu_hat, sigma_hat) ; bug.breaks  # quantile break points

# Observed frequencies
bug.cut = cut(bug_time, breaks = bug.breaks, include.lowest = TRUE, right = TRUE)  # assigning each obs to a bin
bug.table = table(bug.cut) ; bug.table  # observed counts per bin

# Expected frequencies using plnorm
prob = numeric(ncell)
exp.f = numeric(ncell)

for(i in 1:ncell){
  prob[i] = plnorm(bug.breaks[i+1], meanlog = mu_hat, sdlog = sigma_hat) -
    plnorm(bug.breaks[i],   meanlog = mu_hat, sdlog = sigma_hat)  # P(bin i)
  exp.f[i] = prob[i] * n  # expected frequency in bin i
}
exp.f  # = 20 each

# Extract observed frequencies
obs.f = numeric(ncell)
for(i in 1:ncell) obs.f[i] = bug.table[[i]]  # convert table to numeric vector
obs.f

# Chi-square statistic
X2 = sum((obs.f - exp.f)^2 / exp.f) ; X2  # chi-square statistic

# df = (n0. of cells - 1 - no. of estimated parameters)
df = ncell - 1 - 2 ; df  # subtract 2 for estimating mu and sigma^2

# p-value
pval = 1 - pchisq(X2, df) ; pval  # right-tail p-value

# Table with intervals and observed freq
Table = data.frame(
  Interval = names(bug.table),      # interval labels
  Observed = obs.f,                 # observed counts
  Expected = round(exp.f)        # expected counts
)
Table


# Question 2

# Q2a
```

```
# Load the data
app_dur = scan("C:/HWU/Year 3/SEM 2/Stats Model B/cw1/app_session_duration.txt")
n = length(app_dur) ; n

# Histogram and boxplot to visualize the data
hist(app_dur, main = "Histogram of App Session Durations",
    xlab = "Session duration (mins)",
    ylab = "Frequency (No. of sessions)",col = "lavender")

boxplot(app_dur, main = "Boxplot of App Session Durations",
       xlab = "Session duration (mins)",
       col = "lightblue", horizontal = TRUE)

# Numerical summaries for the data
summary(app_dur)
sd(app_dur)      # sample standard deviation
var(app_dur)     # sample variance


# Q2b

B = 10000         # no. of bootstrap resamples
n = length(app_dur)

median.b = numeric(B) # puts empty values so u have space

for(i in 1:B){
  median.b[i] = median(sample(app_dur, size = n, replace = TRUE))  # bootstrap median
}

summary(median.b)  # numerical summaries of bootstrap medians

# The bootstrap distribution plots
# Histogram and boxplots
hist(median.b,
    main = "Histogram of Bootstrap Dist. of the Median",
    xlab = "Bootstrap Median", col = "orange")  # this is the empirical sampling distribution

boxplot(median.b, horizontal = TRUE,
       main = "Boxplot of Bootstrap Medians",
       xlab = "Bootstrap Median", col = "red")

# 95% percentile CI for bootstrap
CI = quantile(median.b, c(0.025, 0.975)) ; CI


# Q2c
# median asymptotic CI

log_app = log(app_dur) # log of data
n = length(log_app)
```

```
m_hat = median(log_app) ; m_hat         # sample median on log-scale
sigma_hat = sd(log_app) ; sigma_hat      # estimate of sigma on log-scale

f_m = 1 / (sigma_hat * sqrt(2*pi))      # since median = mean for Normal

SE_m = 1 / (2 * f_m * sqrt(n)) ; SE_m   # Standard error using general asymptotic formula

# 95% CI for m hat
lower_m = m_hat - 1.96 * SE_m
upper_m = m_hat + 1.96 * SE_m
c(lower_m, upper_m)

# Asymptotic 95% CI for median
CI_median = exp(c(lower_m, upper_m)) ; CI_median


# Q2d

IQR.b = numeric(B)  # storing bootstrap IQRs as empty spaces for now

for(i in 1:B){
  IQR.b[i] = IQR(sample(app_dur, size = n, replace = TRUE))  # bootstrap IQR
}

# 95% percentile CI
CI_IQR = quantile(IQR.b, probs = c(0.025, 0.975)) ; CI_IQR


# Q2e

obs_median = median(app_dur) ; obs_median  # observed median which is the test statistic

mu0 = log(15)  # null value for mu

sigma_hat = sd(log(app_dur))  # use estimated sigma from Q2c

median.b.par = numeric(B)  # storing parametric bootstrap medians as empty spaces for now

for(i in 1:B){
  median.b.par[i] = median(rlnorm(n, meanlog = mu0, sdlog = sigma_hat))  # simulate under H0
}

p_value = mean(median.b.par >= obs_median) ; p_value  # one-sided p-value for the right tail

# Histogram of parametric bootstrap medians
hist(median.b.par,
    main = "Histogram of Parametric Bootstrap Dist. of Median",
    xlab = "Bootstrap Median",
    col = "lightyellow")

# Vertical line for observed median
abline(v = obs_median, col = "red", lwd = 2)  # observed test statistic
```

```
legend("topright",legend = c("Observed median"),
      col = c("red"), lty =1)  # legend for the red line
```

Appendix: Citations

(No date) *F79MB Practical solutions*. Available at: https://canvas.hw.ac.uk/courses/32228/pages/computer-practical-3-sample-solutions?module_item_id=2333299 (Accessed: 22 February 2026).

(No date a) *F79MB Project 1 2025*. Available at: https://canvas.hw.ac.uk/courses/32228/pages/past-project-1?module_item_id=2333325 (Accessed: 22 February 2026).